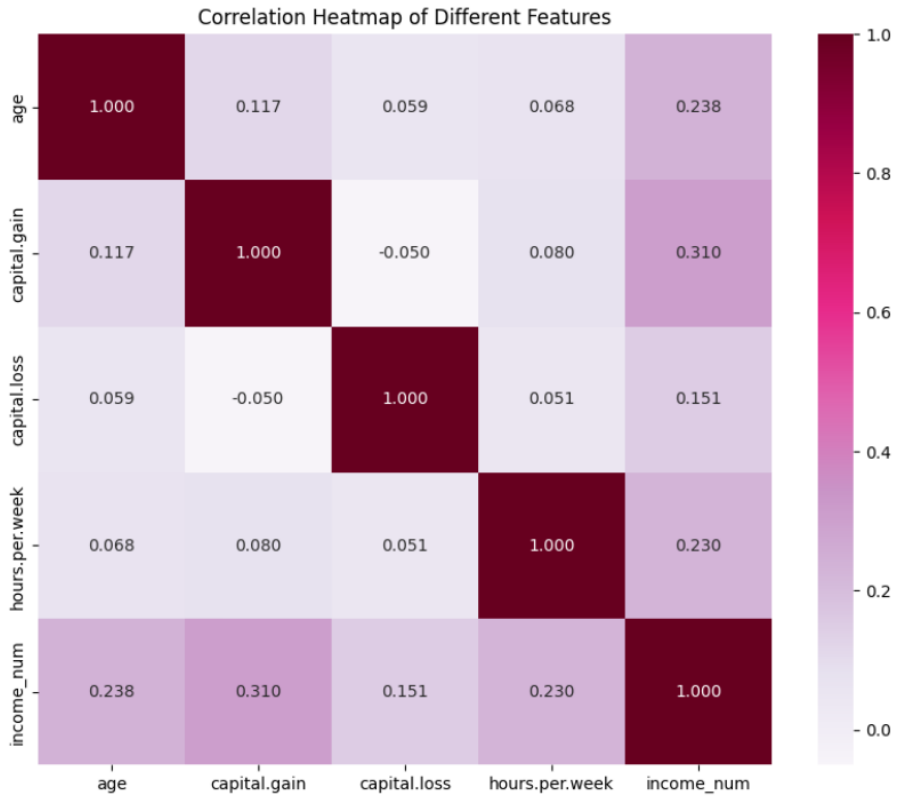
Elisa Herter
DSCI 372 Project 1 Report



1- The correlation heatmap reveals that among the features analyzed, income (income_num) has the strongest positive correlations with capital gain (0.310), age (0.238), and hours per week (0.230), indicating that higher income is generally associated with older age, greater capital gains, and longer working hours. In contrast, capital gain and capital loss show a very weak negative correlation (-0.050), suggesting little to no linear relationship between these two financial variables. Overall, while some features exhibit moderate correlations with income, most other feature pairs have low correlations, highlighting limited linear relationships within the dataset.

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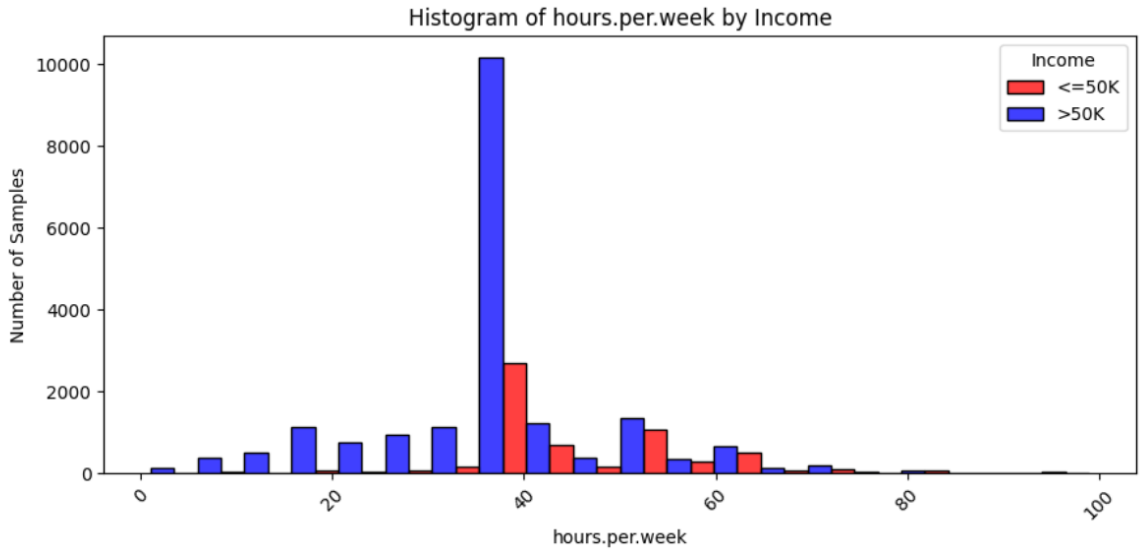

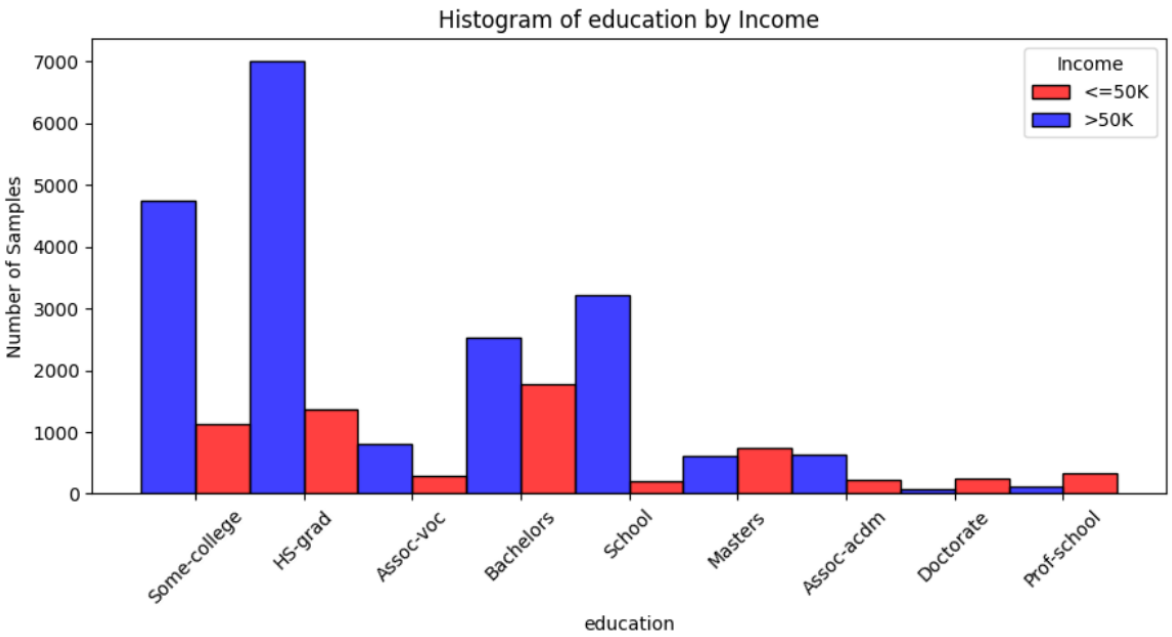

2- The histograms show clear differences by income group. For incomes at or below 50K, common education levels include high school graduate and some college, with typical work hours at 40, 50, and 20 per week, while incomes above 50K cluster around bachelor's degrees and 40 to 50 work hours. A large number of people earning above 50K fall into the high school graduate category, and most higher incomes concentrate around 40 hours per week, which fits the standard full time work schedule in the United States.

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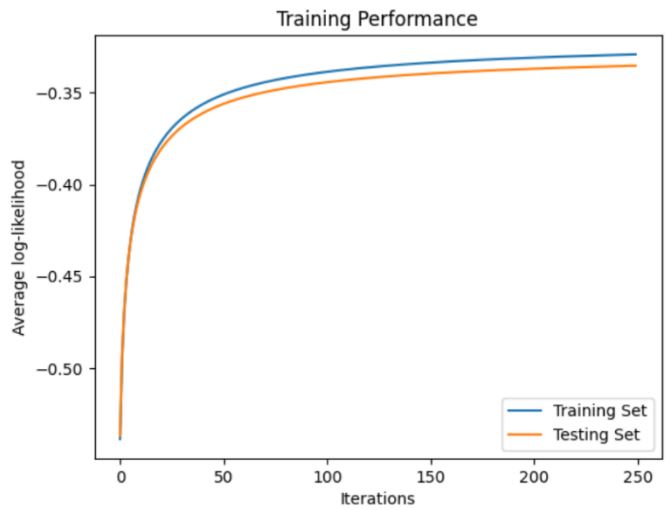
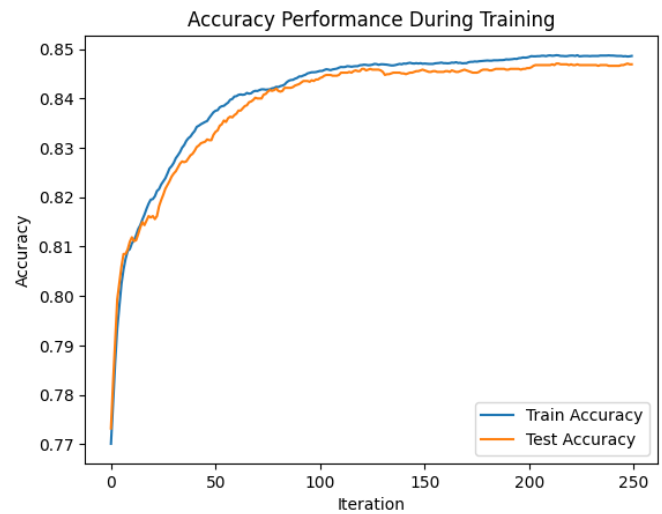
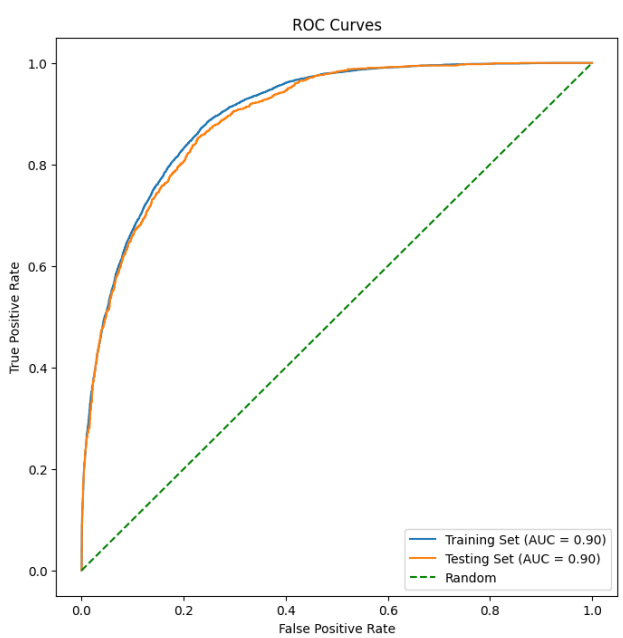




3- The learning rate of 0.05 produces smoother, more stable convergence in both average log-likelihood and accuracy, with training and test curves staying close together and reaching slightly higher final test performance (around 0.845 accuracy). In contrast, the higher learning rate of 0.75 drives faster initial improvement but results in marginally lower final test accuracy (around 0.84) and a slightly larger, though still modest, gap between training and test performance, suggesting a bit more overfitting. Overall, the smaller learning rate appears to offer better generalization and more reliable training dynamics for this model.



4- Both learning rates (0.05 and 0.75) yield essentially identical ROC performance, with training and testing sets achieving the same strong AUC of 0.90 and curves that overlap almost perfectly across the full range of false positive rates. This close alignment indicates excellent discriminative power and very good generalization, with virtually no detectable overfitting in terms of ranking ability. Overall, the choice of learning rate has no meaningful impact on the model's ROC/AUC performance in this case.